ENTERPRISE STORAGE ENGINEERING SOP

Proxmox VE NFS Storage Implementation & Operations Guide

10-Disk Array Provisioning, Server Setup, Cluster Mount, 10G Tuning & Disaster Recovery

Document Version: 1.1-Production
Target Environment: Proxmox VE 8.x / 9.x
Storage Fabric: 10G Optical (192.168.10.0/24)
Storage Media: 10x SAS/SATA/NVMe Disks

1. Overview & Architectural Role of NFS in Proxmox VE

Network File System (NFS) provides a standardized, shared, file-level storage backend across all nodes in a Proxmox VE cluster. While block-level Ceph RBD and local ZFS are ideal for active virtual machine execution, NFS serves as the optimal centralized repository for multi-node operations:

CENTRALIZED BACKUPS (VZDUMP)

- All 3 to 5 cluster nodes write automated VM/LXC backup archives (`.vma.zst`) to a unified NFS target.
- Centralized prune and retention policies across the entire datacenter.

SHARED ISO & TEMPLATES

- Download an ISO or CT template once; instantly available across all cluster nodes.
- Eliminates redundant storage consumption across individual hypervisor hosts.

SHARED VM DISKS & SNIPPETS

- Supports QCOW2/RAW VM disk images.
- Supports zero-copy live VM migration without SAN storage.
- Stores shared Cloud-Init YAML configuration snippets.

2. Supported PVE Content Types on NFS

Proxmox VE automatically creates structured subdirectories inside the NFS export path based on configured content flags:

CONTENT FLAG	SUBDIRECTORY PATH ON NFS	TYPICAL FILE FORMAT	PRIMARY USE CASE IN PRODUCTION
`backup`	`<export>/dump/`	`vzdump-qemu-*.vma.zst`	Automated scheduled VZDump guest backups
`iso`	`<export>/template/iso/`	`*.iso` (e.g. Ubuntu, Windows)	Guest operating system installation media
`vztmpl`	`<export>/template/cache/`	`*.tar.zst, *.tar.gz`	LXC container base OS templates
`images`	`<export>/images/<VMID>/`	`vm-<VMID>-disk-*.qcow2`	Virtual machine virtual disk images
`rootdir`	`<export>/subvol/<CTID>/`	Subvolumes / raw container root	Container root filesystems
`snippets`	`<export>/snippets/`	`*.yaml, *.yml, *.sh`	Cloud-init user-data, vendor-data, and hook scripts

3. Network Fabric & 10G Subnet Routing

To achieve line-rate backup transfer speeds (> 600 MB/s) and eliminate network contention on Corosync cluster heartbeats, NFS traffic must be routed strictly across the dedicated 10G/25G optical network interface (`nic3`):

SUBSYSTEM	NETWORK INTERFACE	SUBNET RANGE	MTU SETTING	DESIGN CONSIDERATION
Management & Corosync	`vmbr0 (eno1 - 1G)`	`10.31.6.0/22`	1500 (Standard)	Dedicated to cluster quorum; zero storage traffic
NFS Storage & Migration	`nic3 (Broadcom 10G)`	`192.168.10.0/24`	**9000 (Jumbo)**	Dedicated 10Gbps optical fabric for Ceph and NFS

IP ADDRESSING BEST PRACTICE: Always specify the NFS Server's `192.168.10.x` IP address (never the management 10.31.6.x address) when mounting NFS in Proxmox. This forces all streaming I/O through the 10GbE optical link.

4. Phase 1: 10-Disk Storage Pool Architecture & Provisioning

Before initializing the NFS daemon, the 10 physical disks must be aggregated into an enterprise-grade resilient volume. The optimal storage layout depends directly on the intended I/O pattern:

4.1. Storage Topology Comparison for 10 Disks

TOPOLOGY	USABLE SPACE (10X 2TB DRIVES)	FAULT TOLERANCE	READ/WRITE PROFILE	PRODUCTION RECOMMENDATION
ZFS RAIDZ2 (Single 10-Disk vdev)	**~80%** (8 Data + 2 Parity) ~16 TB	Survives any **2 disk failures**	High sequential throughput; moderate random IOPS	**Recommended for VZDump Backups, ISOs & Templates**
ZFS Striped Mirrors (RAID 10: 5 pairs)	**50%** (5 Data + 5 Mirror) ~10 TB	Survives **1 disk per pair** (up to 5 total)	Maximum random IOPS; fastest rebuild times	**Recommended for Active VM Disks (QCOW2/RAW)**
ZFS Dual RAIDZ2 (2× 5-disk vdevs)	**60%** (6 Data + 4 Parity) ~12 TB	Survives **2 disks per vdev** (up to 4 total)	Balanced random IOPS and sequential throughput	Mixed workloads (VM disks + VZDump backups)
Linux Software RAID 6 (`mdadm` + XFS)	**~80%** (8 Data + 2 Parity) ~16 TB	Survives any **2 disk failures**	High sequential throughput; standard POSIX filesystem	Non-ZFS environments or systems with low RAM

4.2. Disk Discovery & Persistent Identifier Mapping

Identify disks using persistent serial or WWN paths rather than ephemeral device letters (`/dev/sdX`):

SHELL COMMANDS: IDENTIFY BLOCK DEVICES BY PERSISTENT ID — STORAGE HOST

```
# 1. List block devices and partition tables
$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MODEL,SERIAL
# 2. Inspect persistent device paths (use these in pool creation)
$ ls -la /dev/disk/by-id/
```

4.3. Creating the 10-Disk Storage Pool

OPTION A: ZFS RAIDZ2 (10 DISKS) — RECOMMENDED FOR CENTRALIZED BACKUPS — STORAGE HOST

```
$ zpool create -f -o ashift=12 -O acltype=posixacl -O xattr=sa -O normalization=formD nfs-pool raidz2 \ /dev/disk/by-id/drive1 /dev/disk/by-id/drive2 /dev/disk/by-id/drive3 \ /dev/disk/by-id/drive4 /dev/disk/by-id/drive5 /dev/disk/by-id/drive6 \ /dev/disk/by-id/drive7 /dev/disk/by-id/drive8 /dev/disk/by-id/drive9 \ /dev/disk/by-id/drive10
```

OPTION B: ZFS STRIPED MIRRORS (RAID 10) — RECOMMENDED FOR ACTIVE VM STORAGE — STORAGE HOST

```
$ zpool create -f -o ashift=12 -O acltype=posixacl -O xattr=sa -O normalization=formD nfs-pool \ mirror /dev/disk/by-id/drive1 /dev/disk/by-id/drive2 \ mirror /dev/disk/by-id/drive3 /dev/disk/by-id/drive4 \ mirror /dev/disk/by-id/drive5 /dev/disk/by-id/drive6 \ mirror /dev/disk/by-id/drive7 /dev/disk/by-id/drive8 \ mirror /dev/disk/by-id/drive9 /dev/disk/by-id/drive10
```

4.4. ZFS Dataset Creation & Recordsize Tuning

Set dataset record sizes matching your target workload to maximize disk efficiency and minimize fragmentation:

SHELL COMMANDS: CREATE & TUNE ZFS DATASET — STORAGE HOST

```
# Create dedicated dataset with zstd compression and atime disabled
$ zfs create -o compression=zstd -o atime=off nfs-pool/pve-storage
# For VZDump backups and ISOs, set recordsize to 1M (drastically improves throughput)
$ zfs set recordsize=1M nfs-pool/pve-storage
# Verify dataset properties
$ zfs get compression,recordsize,mountpoint nfs-pool/pve-storage
```

NOTE ON HARDWARE RAID CONTROLLERS: If using a hardware RAID controller (e.g. Dell PERC / MegaRAID), configure a 10-disk RAID-6 virtual disk with 1MB stripe size and Write-Back cache enabled, then format with XFS using `mkfs.xfs -K /dev/sdX`.

5. Phase 2: Deploying & Configuring the NFS Server Daemon

Configure the Linux NFS Kernel Server on the host managing the 10-disk array (e.g. dedicated NAS or designated cluster host):

5.1. Install Packages & Set Storage Permissions

SHELL COMMANDS: INSTALL NFS PACKAGES & CONFIGURE DIRECTORY — NFS HOST

```
# 1. Install kernel NFS server and RPC packages
$ apt update && apt install -y nfs-kernel-server rpcbind
# 2. Configure POSIX permissions for nobody:nogroup
$ chown -R nobody:nogroup /nfs-pool/pve-storage
$ chmod 777 /nfs-pool/pve-storage
```

5.2. Multi-Thread NFS Daemon Tuning for 10 Disks

By default, Linux runs only 8 worker threads for NFS. A 10-disk storage array handling concurrent multi-node backup streams can saturate 8 threads, causing I/O queuing. Increase worker threads to **32 or 64**:

FILE CONFIGURATION: /ETC/NFS.CONF (OR /ETC/DEFAULT/NFS-KERNEL-SERVER) — NFS HOST

```
# Configure 32 parallel worker threads
[nfsd]
threads=32
# Apply thread update:
$ systemctl restart nfs-kernel-server
```

5.3. Configure `/etc/exports` for Proxmox 10G Subnet

Edit `/etc/exports` to restrict access exclusively to the Proxmox 10G network fabric (`192.168.10.0/24`):

FILE CONFIGURATION: /ETC/EXPORTS — NFS SERVER

```
# Proxmox VE Cluster 10G Optical Fabric Export
/nfs-pool/pve-storage 192.168.10.0/24(rw,sync,no_subtree_check,no_root_squash,no_all_squash,sec=sys)
# Reload export configuration live without restarting service:
$ exportfs -arv
exporting 192.168.10.0/24:/nfs-pool/pve-storage
$ systemctl enable --now nfs-kernel-server rpcbind
```

Understanding Export Parameters

OPTION PARAMETER	RECOMMENDED SETTING	ENGINEERING RATIONALE
`rw`	Read-Write	Mandatory for creating backup archives, uploading ISOs, and writing VM images.
`sync`	Synchronous writes	Guarantees write integrity to disk before acknowledging I/O. Prevents corrupted backup chunks.
`no_subtree_check`	Disabled check	Enhances transfer speed and prevents file transfer aborts when renaming backup files.
`no_root_squash`	Allow Root Access	**CRITICAL:** Proxmox backup processes run as root. Without this, PVE cannot create backup archives or write image metadata.
`sec=sys`	Standard UNIX Auth	Standard UID/GID matching for secure, fast internal cluster access over private VLANs.

5.4. Firewall Configuration (Port 2049 & 111)

SHELL COMMANDS: OPEN NFS FIREWALL PORTS ON UFW / IPTABLES — NFS SERVER

```
# Allow NFS v4 (TCP 2049) and RPC (TCP/UDP 111) from 10G subnet only
$ ufw allow from 192.168.10.0/24 to any port 2049 proto tcp
$ ufw allow from 192.168.10.0/24 to any port 111 proto tcp
$ ufw allow from 192.168.10.0/24 to any port 111 proto udp
```

6. Phase 3: Mounting NFS Storage in Proxmox VE

Step 6.1: Verify Remote NFS Reachability from Proxmox Nodes

```
SHELL COMMAND: QUERY NFS EXPORTS FROM PVE NODE                                TARGET: PROX22
prox22$ showmount -e 192.168.10.50 # (Replace with your NFS Server 10G IP)
Export list for 192.168.10.50:
/nfs-pool/pve-storage 192.168.10.0/24
```

Step 6.2: Method 1 — Adding NFS via Proxmox CLI (`pvesm`)

Adding NFS storage via `pvesm add nfs` automatically propagates cluster-wide to all nodes via `/etc/pve/storage.cfg`:

```
SHELL COMMAND: ADD NFS STORAGE VIA PVE STORAGE MANAGER                        TARGET: PROX22
$ pvesm add nfs nfs-10disk-storage \ --server 192.168.10.50 \ --export /nfs-pool/pve-storage \ --content backup,iso,vztmpl,images \
--options "vers=4.2,tcp,nconnect=8,timeo=600,retrans=2" \ --prune-backups "keep-last=3,keep-daily=7,keep-weekly=4,keep-monthly=12"
# Verify storage status across all cluster nodes:
$ pvesm status
Name                  Type     Status           Total            Used       Available        %
ceph                   rbd     active     76303248384     10821304320     65481944064   14.18%
zfs-ha              zfspool     active     91100344320      8589934592     82510409728    9.43%
nfs-10disk-storage      nfs     active    160000000000     12000000000    148000000000    7.50%
```

Step 6.3: Method 2 — Adding NFS via Proxmox Web GUI

1. Log in to the Proxmox Web GUI (`https://10.31.6.172:8006`).
2. In the left navigation tree, select **Datacenter** → **Storage** → Click **Add** → Select **NFS**.
3. Enter parameters:
   - **ID:** `nfs-10disk-storage`
   - **Server:** `192.168.10.50` (NFS Server 10G Optical IP)
   - **Export:** Select `/nfs-pool/pve-storage` from dropdown
   - **Content:** Select `VZDump backup file`, `ISO image`, `Container template`, `Disk image`
   - **Nodes:** Leave empty (All nodes) for cluster-wide mount
4. Click **Add**. Proxmox synchronously mounts the NFS volume across all nodes within seconds.

7. Phase 4: 10GbE High-Performance Tuning Parameters

Default Linux NFS uses a single TCP stream. On high-speed 10GbE optical interconnects, configure connection multiplexing (`nconnect=8`) and 1MB buffers:

MOUNT OPTION	RECOMMENDED VALUE	PERFORMANCE BENEFIT ON 10G OPTICAL INTERCONNECT
`vers=4.2`	NFSv4.2	Enables Server-Side Copy (SSC) and sparse file allocation for ultra-fast VM cloning.
`nconnect=8`	8 TCP Streams	Multiplexes RPC requests across 8 parallel TCP connections, saturating 10GbE bandwidth.
`rsize=1048576`	1 MB Read Buffer	Optimizes sequential restore throughput from NFS to local Ceph/ZFS pools.
`wsize=1048576`	1 MB Write Buffer	Optimizes VZDump streaming backup throughput (> 600 MB/s sustained).
`timeo=600`	60 Seconds Timeout	Prevents premature timeout during high-throughput parallel backup commits.

8. Phase 5: Automated Scheduled Backups (VZDump) to NFS

Step 8.1: Schedule Daily Backups via Proxmox Datacenter

1. In PVE GUI: **Datacenter** ➔ **Backup** ➔ Click **Add**.
2. Configure parameters:
   - **Node:** All (Entire Cluster) | **Storage:** `nfs-10disk-storage`
   - **Schedule:** `01:00` (Nightly at 1:00 AM) | **Selection Mode:** `All guests`
   - **Mode:** `Snapshot` (Zero downtime) | **Compression:** `Zstandard (zstd)`
   - **Retention/Prune:** Keep last 3, Keep daily 7, Keep weekly 4, Keep monthly 12
3. Click **Create**. Automatically registered in `/etc/pve/vzdump.cron`.

Step 8.2: Test Manual Backup Execution via CLI

```
SHELL COMMAND: TEST SNAPSHOT BACKUP OF VM 9001 TO NFS                                TARGET: PROX22
$ vzdump 9001 --storage nfs-10disk-storage --mode snapshot --compress zstd --notes-template "{{guestname}} (Weekly Full)"
INFO: starting new backup job: vzdump 9001 --mode snapshot --storage nfs-10disk-storage --compress zstd
INFO: Starting Backup of VM 9001 (qemu)
INFO: status = running
INFO: backup mode: snapshot
INFO: creating VZDump architecture archive to 'nfs-10disk-storage:backup/vzdump-qemu-9001-2026_09_18-17_15_00.vma.zst'
INFO: transferring 4.00 GiB [==========================>] 100% in 7s (average 585.1 MiB/s)
INFO: archive and directory size: 1.42 GiB
INFO: Finished Backup of VM 9001 (00:00:11)
INFO: Backup job finished successfully
```

9. Phase 6: Maintenance & Troubleshooting Playbooks

9.1. 10-Disk Pool Maintenance & Disk Replacement

```
PLAYBOOK: MONITOR HEALTH, SCRUB & REPLACE FAILED DISK                                STORAGE HOST
# 1. Check array status and error counters
$ zpool status nfs-pool
# 2. Initiate monthly background data scrub
$ zpool scrub nfs-pool
# 3. Disk Replacement Procedure:
$ zpool offline nfs-pool /dev/disk/by-id/old-drive-serial
# (Physically swap drive in chassis hot-swap bay)
$ zpool replace nfs-pool /dev/disk/by-id/old-drive-serial /dev/disk/by-id/new-drive-serial
```

9.2. Playbook A: NFS Storage Shows Question Mark `(?)` Icon in Web GUI

```
PLAYBOOK: DIAGNOSE AND RE-ARM INACTIVE NFS STORAGE                                   CLUSTER TERMINAL
# 1. Test basic IP and port connectivity over 10G optical interface
$ ping -c 3 192.168.10.50 && nc -zv 192.168.10.50 2049
# 2. Check if export is broadcasted
$ showmount -e 192.168.10.50
# 3. Trigger manual storage scan and mount refresh
$ pvesm set nfs-10disk-storage --disable 1 && sleep 2 && pvesm set nfs-10disk-storage --disable 0
$ pvesm status
```

9.3. Playbook B: Stale File Handle (`ESTALE`) Resolution

```
PLAYBOOK: CLEAR STALE MOUNTPOINTS VIA LAZY UNMOUNT                                   TARGET NODE
# 1. Force detach stale mount from namespace
$ umount -l /mnt/pve/nfs-10disk-storage
# 2. Refresh storage status to trigger clean reconnection
$ pvesm status
```

9.4. Playbook C: Node Boot Hangs Waiting for Unreachable NFS Server

If the NFS server is powered down, Proxmox nodes take up to 90 seconds per mount to boot due to blocking systemd mount units. Enable non-blocking systemd automounting:

```
PLAYBOOK: ENABLE NON-BLOCKING AUTO-MOUNTING (X-SYSTEMD.AUTOMOUNT)  TARGET NODE
# Update PVE storage options to include nofail and background automounting:
$ pvesm set nfs-10disk-storage --options "vers=4.2,tcp,nconnect=8,timeo=600,retrans=2,_netdev,nofail,x-systemd.automount,x-systemd.idle-timeout=60"
```

9.5. Playbook D: Permission Denied (`EACCES` / `EPERM`) on Writing Backups

```
PLAYBOOK: FIX UID/GID SQUASH & DIRECTORY OWNERSHIP  NFS SERVER
# 1. Verify /etc/exports contains no_root_squash
$ cat /etc/exports | grep pve-storage
# 2. Reset filesystem permissions and reload export table
$ chmod 777 /nfs-pool/pve-storage && exportfs -arv
```

10. Daily Operational Cheat Sheet for NFS Storage

ADMINISTRATIVE TASK	EXACT SHELL COMMAND	HEALTHY EXPECTED OUTPUT
Inspect 10-Disk Pool Health	`zpool status nfs-pool`	`state: ONLINE`, 0 errors across all 10 drives
Check Storage Health on PVE	`pvesm status -storage nfs-10disk-storage`	`Status: active`
List Existing Backups	`pvesm list nfs-10disk-storage --content backup`	List of `.vma.zst` backup archives with sizes
List Available ISOs	`pvesm list nfs-10disk-storage --content iso`	List of available OS installation ISOs
Download ISO Directly to NFS	`pveam download nfs-10disk-storage <TEMPLATE_NAME>`	Downloaded and immediately visible cluster-wide
Check Mount Performance	`nfsstat -m /mnt/pve/nfs-10disk-storage`	Displays active options (`vers=4.2`, `rsize=1048576`, `nconnect=8`)
Benchmark Direct Write Speed	`dd if=/dev/zero of=/mnt/pve/nfs-10disk-storage/test.img bs=1M count=1000 oflag=direct`	Speed > `550 MB/s` over 10G optical fabric

11. Engineering Document Sign-Off

Document Author:
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Signature: _____

Storage Architecture Review:
10-Disk Array + NFSv4.2 10G Certified
VZDump Multi-Node Retention
Signature: _____

Management Approval:
[ ] Approved for Production
[ ] Approved for Staging
Signature: _____